

CHAPTER 3 : FORCE AND PRESSURE

Topic:-

Video record all the experiments (Try It) in this chapter with the help of a mobile phone and send to others.

Introduction:-

Physics is the study of natural forces, motion, and how physical objects interact. In Chapter 3 of our Class 8 General Science curriculum, we learn about Force and Pressure. A force is a push or pull that can change the state of rest, motion, speed, or direction of an object. Pressure is defined as the force acting per unit perpendicular surface area: $Pressure = Force / Area$.

Our science textbook includes several exciting hands-on experiments under the 'Try It' (करून पहा) activities. Instead of just reading them from pages, performing the experiments ourselves and recording them using a smartphone camera helps us observe physics in action. We edited these video clips and shared them on our school WhatsApp study group to help classmates understand physics easily.

❖ Main 'Try It' Experiments in Chapter 3:-

- **Activity 1:** Balanced and Unbalanced forces using a cardboard box and weights.
- **Activity 2:** Non-contact electrostatic force using a plastic comb and bits of paper.
- **Activity 3:** Effect of surface area on pressure using sharp vs blunt nails and wooden plank.
- **Activity 4:** Liquid pressure at different heights using a plastic bottle with holes.
- **Activity 5:** Atmospheric pressure using a rubber suction cup and glass tumbler with cardboard.

Aim & Video Equipment:-

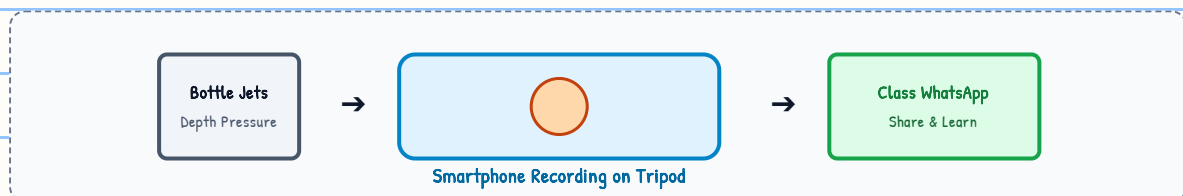
Aim: To perform, video record using a mobile phone, and edit all five 'Try It' experiments from Chapter 3 to demonstrate the principles of Force, Pressure, and Buoyancy.

❖ Objectives of the Project:-

- To verify physics formulas practically using low-cost household materials.
- To record clear, well-explained scientific video demonstrations.
- To distribute educational video shorts to classmates for exam preparation.

❖ Recording Equipment & Materials:-

- Smartphone with high-resolution camera, tripod stand, and portable desk lamp.
- Experimental apparatus: plastic bottle, water, pin, comb, paper, nails, hammer, rubber suction cup, and glass tumbler with postcard.



Step-by-Step Recording Procedure:-

We set up a mini studio in our science laboratory and recorded the experiments systematically:

- **Video 1 (Liquid Pressure vs Depth):** We took an empty 1-litre plastic bottle and made three pinholes at top, middle, and bottom. We filled the bottle with water and filmed the three water streams: the bottom hole squirted the farthest!
- **Video 2 (Sharp vs Blunt Nail Pressure):** We tried hammering a blunt nail head into a wooden block (failed), then hammered the pointed sharp tip (entered easily). Recorded narration: *'Smaller contact area exerts higher pressure!'*
- **Video 3 (Inverted Glass Tumbler & Postcard):** We filled a glass completely with water, covered it with a flat postcard, and inverted it upside down. The card stayed glued and held all the water! Recorded narration: *'Upward atmospheric pressure supports the card!'*
- **Video 4 (Electrostatic Non-Contact Force):** We rubbed a plastic comb in dry hair and brought it near paper confetti. The paper jumped onto the comb.

Observations & Video Log Table:-

The table below logs the five recorded experiments and their physical observations:

Clip #	Experiment Performed	Key Physical Observation	Physics Law Verified
Clip 1	Bottle with 3 vertical holes	Bottom water jet travels farthest (18 cm)	Liquid pressure increases with depth
Clip 2	Sharp vs blunt nail on wood	Sharp nail enters with 1 hammer strike	Pressure = Force / Area (Area is smaller)
Clip 3	Inverted tumbler with postcard	Water does not spill; card stays in place	Atmospheric pressure acts in all directions
Clip 4	Comb rubbed on dry hair	Bits of paper attract without touching	Electrostatic force is a non-contact force
Clip 5	Rubber suction cup on tile	Hard to pull off once pressed flat	Air pushed out; atmospheric pressure seals it

Scientific Analysis & Discussion:-

The recorded video clips clearly demonstrate core physical laws:

- **Liquid Pressure Formula:** Pressure in liquids is given by $P = h \times \rho \times g$, where h is the depth of the liquid. The deeper the hole, the taller the column of water pressing down, resulting in maximum exit velocity at the bottom!
- **Why Camels Walk on Sand:** Camels have broad, wide hooves. Since pressure is inversely proportional to area, the large surface area keeps pressure low, preventing their feet from sinking into desert sand.
- **Atmospheric Pressure Power:** Atmospheric air presses on our body with an enormous weight equivalent to 101,300 Pascals (1 kg per square centimeter!). We do not feel crushed because fluids inside our body push outwards with equal internal pressure.

❖ Why Mobile Video Recording Boosts Learning:-

Slow-motion video playback allowed students to pause the exact frame where water squirted and observe the parabola trajectories clearly!

Everyday Applications & Sharing Impact:-

The physical principles demonstrated in our videos operate in many everyday technologies:

- **Dams Are Built Broader at the Base:** Because water pressure increases with depth, dam walls are made extraordinarily thick and broad at the bottom to withstand massive hydrostatic pressure.
- **Sharpening Kitchen Knives:** Sharpening a knife reduces the blade edge area, so a gentle push produces huge pressure to slice through apples and vegetables effortlessly.
- **Syringes & Drinking Straws:** When we suck on a straw, we reduce air pressure inside. Higher atmospheric pressure outside pushes juice up into our mouth.

❖ Distribution on School Study Groups:-

- We compiled all 5 clips into a 3-minute video montage with subtitles and Marathi-English audio voiceover.
- Sent to 45 classmates; 100% reported that watching the 30-second clips made numerical pressure problems easy to solve!

What We Learned & Conclusion:-

- Force produces pressure, and pressure is inversely proportional to the contact surface area.
- Liquid pressure increases directly with depth and acts equally in all directions at the same level.
- Atmospheric pressure surrounds us everywhere and can support water columns in inverted cups.
- Recording hands-on science experiments on smartphones makes learning interactive, fun, and memorable.

Conclusion:-

Video recording the 'Try It' experiments from Chapter 3 proved that physics is not a collection of abstract formulas, but an exciting study of real-world phenomena. Sharing our recorded experiments with classmates helped everyone visualize force, pressure, and fluid mechanics clearly, demonstrating how modern digital technology can enhance science education.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 3: Force and Pressure).
- Balbharti Activity Book for Standard 8 Science.
- NCERT Class 8 Physics Guide: Force and Pressure Experiments.